

Exam Plan.

Name: BrightLearn Data Warehouse Build

1. Bright Learn operates a customer loyalty programme with three tiers: Bronze - Silver - Gold. Which rewards repeat purchases.

2. We have 5 stores

Sandton (Gauteng)	} Province & City
Cape Town (Western Cape)	
Umhlanga (KwaZulu-Natal)	
Pretoria (Gauteng)	
Gqeberha (Eastern Cape)	

Each store runs its own Point-of-Sale (POS) system, which records all customer transactions. We have different types

NOTE: It means we are dealing with physical stores and POS

POS → The hardware and software businesses use to process transactions, accept payments, and manage daily operations like inventory and sales tracking.

When a customer makes a purchase, the POS system calculates the total (including taxes and discounts) and processes their payment (cash, card, digital wallet). Once the transaction is authorized, the system records the sale and automatically updates your store's inventory.

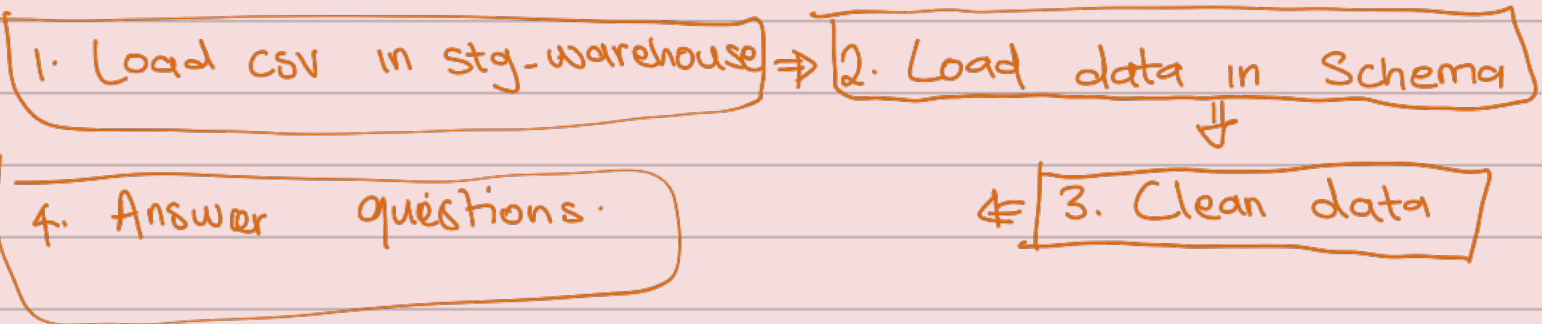
Problem: POS system exports all sales data as a single flat file. This

file contains all transaction details, customer details, product details, and store details mixed together in one table - with

no normalisation, no consistent formatting, and known data quality issues that have accumulated over time.

Solution: Functional data warehouse to make data-driven decisions about inventory, customer retention, or store performance

SSIS ETL Plan



Technical Tools

- * SQL Server
- * SSMS
- * SSIS
- * Github
- * CSV
- * SQL

Assessment Weight

Database Design (Schema & Normalisation) $\Rightarrow$ 20%.

SSIS ETL $\Rightarrow$ 30%.

Data Quality $\Rightarrow$ 15%.

Queries $\Rightarrow$ 25%.

Github $\Rightarrow$ 5%.

Code $\Rightarrow$ 5%.

Due date: 21 July 2026 23:59

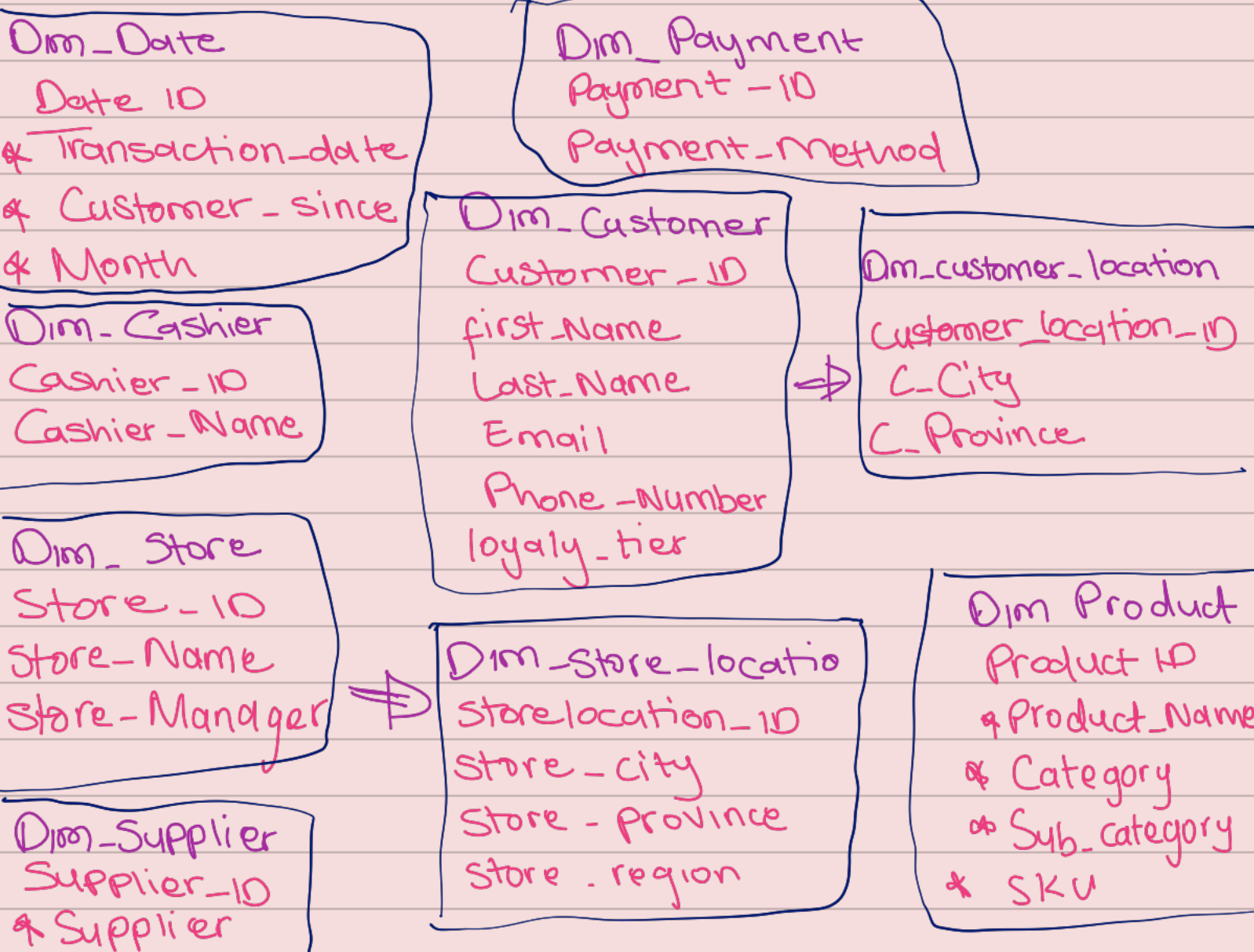
Dataset Inspection.

SKU - Category - Sub-Category

SKU $\Rightarrow$ Stock Keeping Unit "Skew" $\Rightarrow$ Unique ID

Column	Problem
transaction_date	Inconsistent date format
transaction_amount	Column in text instead of number
transaction_discount	Column in text instead of number
Unit-Price	"
Cost-Price	"
line_amount	
Customer_email	Blank rows
Customer-Phone	9 digit number add a '0' at beginning since its SA ¹²⁷

Star Schema Plan.



Fact - Sales

Customer - ID

Customer - location - ID

Product - ID

Store - location - ID

Store - ID

Supplier - ID

Cashier - ID

Date - ID

Payment - ID

transaction_amount

transaction_discount

Unit_Price

Cost_Price

quantity

line_amount

Stock-on-hand $\Rightarrow$ Total quantity of stock in Storage location

reorder_threshold $\Rightarrow$ (reorder Point)

line_amount $\Rightarrow$ represent cost of that individual line item before any overarching invoice-level discounts or taxes are applied
 $\text{Line_Amount} = \text{Unit_price} \times \text{Quantity}$

* If multiple quantities are purchased with an applicable line discount: $\text{Line_Amount} = \text{Unit_price} \times \text{Quantity} \times (1 - \text{discount})$

* reorder_threshold $\Rightarrow$ Specific inventory level that triggers the need to buy more stock

* When stock-on-hand drops to this exact number, the retailer must place new order to restock before running

out of items WHERE Stock-on-hand = reorder-threshold

AND Stock-on-hand < reorder-threshold

* This to find items that need reorder and those that are overdue for immediate purchase.

* I need to find the percentage of the discount and how its affecting revenue.

* Calculate revenue / Profit.